



Mortgage Pricer — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

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1 Test Results — Mortgage Pricer (MKM-MP-001)

Tests run on **04 April 2026 at 18:43:46**.

Table 1: Test Results Summary — Mortgage Pricer (MKM-MP-001)

Total Tests	67
Passed	67
Failed	0
Skipped	0
Duration	0.05s

Table 2: Detailed Test Results — Mortgage Pricer

Test Description	Acceptance Criteria	Result	Time
Default ids assigned	<code>results[0]["mortgage_id"] == "MORTGAGE_0";</code> <code>results[0]["property_id"] == "PROPERTY_0"</code>	Pass	0.001s
Error isolation	<code>len(results) == 3;</code> "error" not in <code>results[0]</code> (+2 more)	Pass	0.002s
Prices multiple mortgages	<code>len(results) == 2;</code> <code>results[0]["mortgage_id"] == "M1"</code> (+1 more)	Pass	0.002s
Custom tax rate	<code>MortgagePricer(tax_rate=0.40).tax_rate == 0.40</code>	Pass	0.000s
Default tax rate	<code>MortgagePricer().tax_rate == 0.20</code>	Pass	0.000s
Custom tax rate	<code>spread_high_tax > spread_low_tax</code>	Pass	0.000s
Floor at 10bps	<code>spread >= 0.001</code>	Pass	0.000s
Higher affordability higher spread	<code>spread_low_income > spread_high_income</code>	Pass	0.000s
Negative income fallback	<code>spread == 0.15</code>	Pass	0.000s
Spread bounded by schedule	<code>spread >= 0.001; spread < 0.05</code>	Pass	0.000s
Zero income fallback	<code>spread == 0.15</code>	Pass	0.000s
debug=True in the credit spread calculation.	<code>"credit_spread" in result</code>	Pass	0.001s
debug=True exercises logging branches without error.	<code>result["mortgage_value"] > 0</code>	Pass	0.001s
Zero income with debug=True exercises zero-income debug branch.	<code>result["credit_spread"] > 0</code>	Pass	0.001s
flood_risk_category=None with debug=True.	<code>result["mortgage_value"] > 0</code>	Pass	0.001s

Test Description	Acceptance Criteria	Result	Time
When interest_rate=0, monthly_rate==0 → loan_amount/n_periods formula.	result["mortgage_value"] > 0	Pass	0.001s
Batch passes flood risk	results[0]["flood_risk_factor"] == 1.40; results[1]["flood_risk_factor"] == 1.00 (+1 more)	Pass	0.002s
Case insensitive	MortgagePricer.calculate_flood_risk_impact("high") == ...; MortgagePricer.calculate_flood_risk_impact("VERY LOW") == ... (+1 more)	Pass	0.000s
Flood factor in result	result["flood_risk_factor"] == 1.20	Pass	0.001s
Flood risk increases spread	high["credit_spread"] > low["credit_spread"]; high["flood_risk_factor"] > low["flood_risk_factor"] (+1 more)	Pass	0.001s
Flood risk multipliers[high-1.4] [category='High', expected=1.4]	MortgagePricer.calculate_flood_risk_impact("High") == e...	Pass	0.000s
Flood risk multipliers[low-1.05] [category='Low', expected=1.05]	MortgagePricer.calculate_flood_risk_impact("Low") == e...	Pass	0.000s
Flood risk multipliers[medium-1.2] [category='Medium', expected=1.2]	MortgagePricer.calculate_flood_risk_impact("Medium") == e...	Pass	0.000s
Flood risk multipliers[very high-1.75] [category='Very High', expected=1.75]	MortgagePricer.calculate_flood_risk_impact("Very High") == e...	Pass	0.000s
Flood risk multipliers[very low-1.0] [category='Very Low', expected=1.0]	MortgagePricer.calculate_flood_risk_impact("Very Low") == e...	Pass	0.000s
No flood risk defaults to unity	result["flood_risk_factor"] == 1.0	Pass	0.001s
None returns unity	MortgagePricer.calculate_flood_risk_impact("None") == 1.0	Pass	0.000s
Unknown category returns unity	MortgagePricer.calculate_flood_risk_impact("Unknown") == 1.0; MortgagePricer.calculate_flood_risk_impact("Extreme") == 1.0	Pass	0.000s
Ltv thresholds[0.5-1.0] [ltv=0.5, expected=1.0]	pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...	Pass	0.000s
Ltv thresholds[0.8-1.0] [ltv=0.8, expected=1.0]	pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...	Pass	0.000s
Ltv thresholds[0.85-1.1] [ltv=0.85, expected=1.1]	pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Ltv thresholds[0.9-1.1] [ltv=0.9, expected=1.1]	<code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>	Pass	0.000s
Ltv thresholds[0.91-1.3] [ltv=0.91, expected=1.3]	<code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>	Pass	0.000s
Ltv thresholds[0.95-1.3] [ltv=0.95, expected=1.3]	<code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>	Pass	0.000s
Ltv thresholds[0.96-1.5] [ltv=0.96, expected=1.5]	<code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>	Pass	0.000s
Ltv thresholds[1.0-1.5] [ltv=1.0, expected=1.5]	<code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>	Pass	0.000s
Zero property value	<code>pricer.calculate_loan_to_value_impact(100.0, 0) == 1.5</code>	Pass	0.000s
Higher rate higher payment	<code>high > low</code>	Pass	0.000s
Shorter term higher payment	<code>short > long</code>	Pass	0.000s
£200k at 3.5% over 25yr £1,001.25/month.	<code>995 < payment < 1010</code>	Pass	0.000s
Zero-rate degenerate case: $M = P / n$.	<code>payment == pytest.approx(1000.0)</code>	Pass	0.000s
Each mortgage has property id	<code>mapping.get("property_id") is not None</code>	Pass	0.003s
Processing stats	<code>stats["successful_mortgages"] == 5;</code> <code>stats["failed_mortgages"] == 0</code>	Pass	0.003s
Output file exists	<code>result["file_path"].exists()</code>	Pass	0.002s
Output file is valid json	<code>isinstance(data, dict)</code>	Pass	0.004s
Generate returns expected keys	<code>"data" in result;</code> <code>"file_path" in result (+1 more)</code>	Pass	0.003s
Mortgage count matches properties	<code>len(result["data"]["mortgages"]) == 5</code>	Pass	0.003s
Mortgage id format	<code>re.match(r"~MORT-[a-f0-9]{8}\$", mid)</code>	Pass	0.004s
Mortgage ids unique	<code>len(ids) == len(set(ids))</code>	Pass	0.003s
All errors returns error	<code>"error" in calculate_portfolio_metrics(results)</code>	Pass	0.000s
High risk count	<code>calculate_portfolio_metrics(results)["high_risk_mortgages"]</code>	Pass	0.000s
Mixed results exclude errors	<code>metrics["total_mortgages"] == 1;</code> <code>metrics["error_count"] == 1</code>	Pass	0.001s
Valid results	<code>metrics["total_mortgages"] == 2;</code> <code>metrics["error_count"] == 0 (+3 more)</code>	Pass	0.002s
Balance converges to zero	<code>result["outstanding_balance"][-1] < 1.0</code>	Pass	0.001s
Expected losses non negative	<code>all(e1 >= 0 for e1 in result["expected_losses"])</code>	Pass	0.001s

Test Description	Acceptance Criteria	Result	Time
Fair value less than par	<code>result["mortgage_value"] < base.mortgage_params["loan_amo...; result["discount_to_par"] > 0 (+1 more)</code>	Pass	0.001s
Hazard rates positive	<code>result["hazard_rates"][0] == 0.0; all(result["hazard_rates"][i] > 0 for i in range(1, len(r...</code>	Pass	0.001s
High ltv increases spread	<code>high_ltv["credit_spread"] > low_ltv["credit_spread"]; high_ltv["ltv_factor"] > low_ltv["ltv_factor"]</code>	Pass	0.001s
Higher haircut more loss	<code>high["pv_losses"] > low["pv_losses"]</code>	Pass	0.001s
Input validation clamps	<code>result["mortgage_value"] > 0</code>	Pass	0.000s
Ltv factor in result	<code>"ltv_factor" in result; result["ltv_factor"] == 1.0</code>	Pass	0.001s
Raises on invalid types	—	Pass	0.000s
Recovery haircut zero low ltv	<code>all(lgd == 0 for lgd in result["lgds"])</code>	Pass	0.001s
Survival first period less than one	<code>result["survival_probs"][1] < 1.0</code>	Pass	0.001s
Survival monotonically decreasing	<code>surv[i] <= surv[i - 1]</code>	Pass	0.001s
Positive rate exceeds principal	<code>cost > 200.000</code>	Pass	0.000s
Zero rate equals principal	<code>cost == pytest.approx(100_000.0)</code>	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
05-Mar-2026	1.0	Initial test suite (65 tests)	D. Kelly
07-Mar-2026	1.1	62 test(s) added; 65 test(s) removed (total: 62)	D. Kelly
04-Apr-2026	1.2	5 test(s) added (total: 67)	D. Kelly